SIMATS ENGINEERING
ASSESSMENT TOOL 1 - High (INDUSTRY PROBLEM SOLVING TASKS)

COURSE CODE: CSA0921 **COURSE NAME: Programming in Java**

COURSE OUTCOME COVERED

CO4: *Design and develop GUI-based user interfaces using Abstract Window Toolkit, Swing, and Applets by implementing event handling, layout managers, AWT controls, and menus. (BL3,BL5)*

Assessment Tool Weightage: 10%

QUESTIONS: 5 **Total Marks: 50**

ANNEXURE A

INDUSTRY PROBLEM SOLVING TASKS

1. Smart Attendance System (BTL: BL3 – Apply)

Develop a GUI-based attendance system for an organization using Swing components. The system should allow marking attendance, viewing reports, and updating records. Use event handling to process user inputs and actions. Implement validation to prevent duplicate entries and incorrect data. Apply suitable layout managers for structured display. Analyze system responsiveness during bulk data entry and suggest enhancements for better performance and usability.

Code of Conduct:

*I **BHEESHMA L / 192511332** certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.*
I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:
BHEESHMA L

RUBRICS FOR CALCULATION:

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Understanding of Industry Problem	20%	Comprehensive understanding of real-world problem and constraints	Good understanding with minor gaps in requirements	Basic understanding; some requirements unclear	Poor understanding; misinterprets problem context
Application of Engineering Concepts	25%	Applies advanced and relevant concepts effectively to solve the problem	Applies appropriate concepts with minor errors	Limited application of concepts	Incorrect or no application of concepts
Solution Design	25%	Innovative, practical, and feasible solution aligned with industry standards	Logical and feasible solution with minor gaps	Basic solution with limited feasibility	Unclear or impractical solution
Use of Tools	15%	Effective use of modern tools, technologies, and real data	Appropriate use of tools with correct implementation	Limited or inconsistent use of tools	Minimal or incorrect use of tools
Analysis	10%	Thorough analysis with validated results and performance evaluation	Good analysis with reasonable validation	Basic analysis with limited validation	Poor or no analysis
Professionalism	5%	Highly professional, well-documented, and clearly presented work	Good documentation and presentation	Basic documentation with some gaps	Poor documentation and presentation

Smart Attendance System

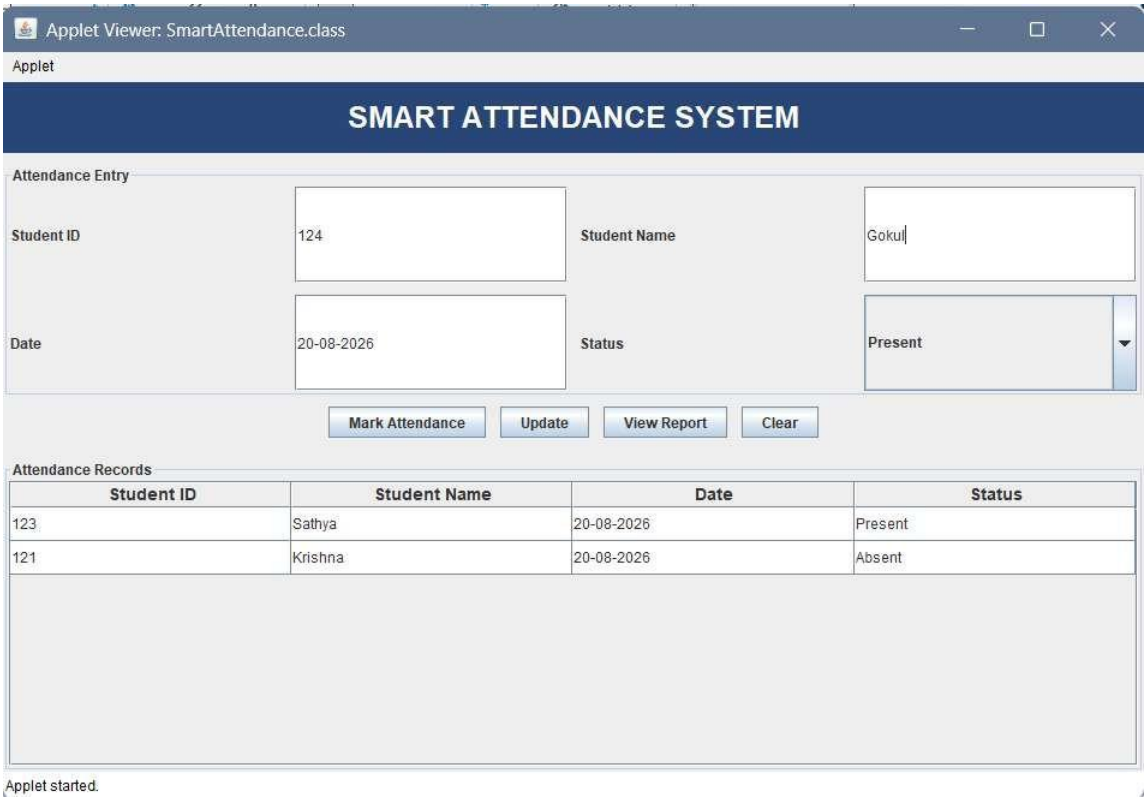
Source Code:

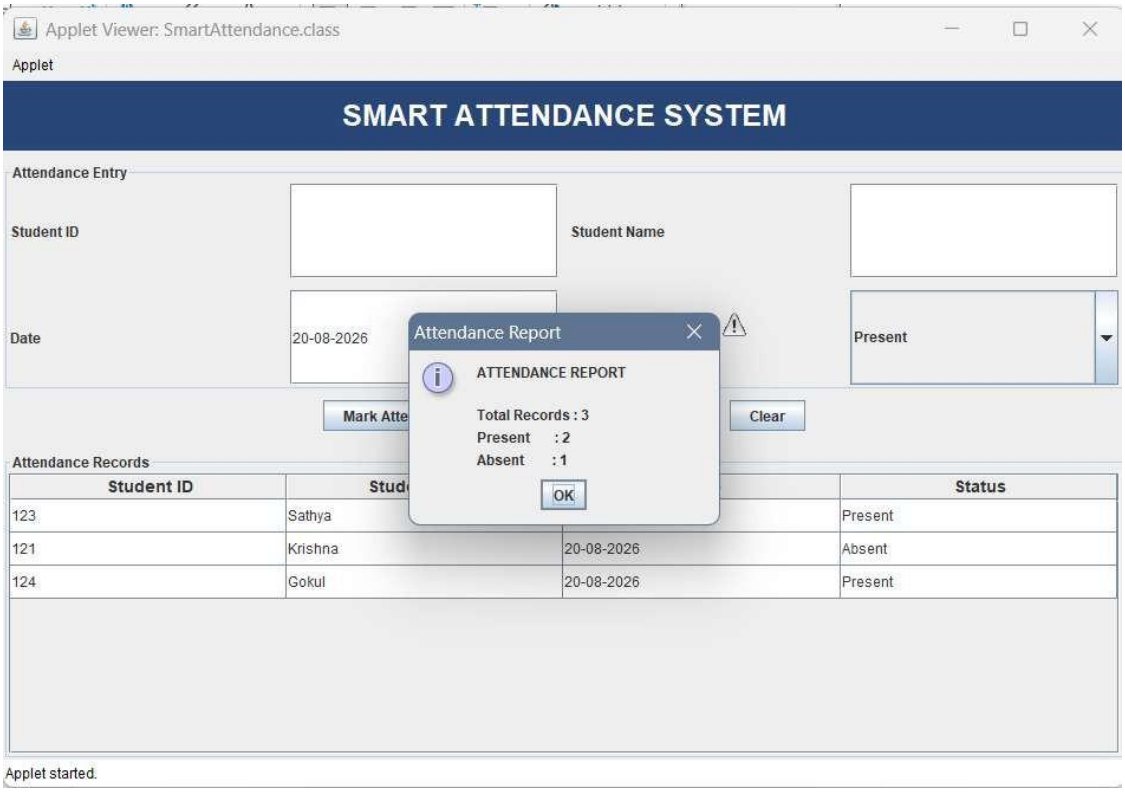
```java
import javax.swing.*;
import javax.swing.table.DefaultTableModel;
import java.awt.*;
import java.awt.event.*;
import java.util.HashSet;

public class SmartAttendance extends JApplet implements ActionListener {

    // Input components
    JTextField nameField, idField, dateField;
    JComboBox<String> statusBox;

    // Table and buttons
    JTable table;
    DefaultTableModel model;
    JButton markBtn, updateBtn, reportBtn, clearBtn;

    // Used to prevent duplicate entries
    HashSet<String> attendanceIds = new HashSet<String>();

    // Main initialization
    public void init() {
        setSize(950, 600);
        setLayout(new BorderLayout(10, 10));

        // ---------- HEADER ----------
        JLabel title = new JLabel(
            "SMART ATTENDANCE SYSTEM",
            SwingConstants.CENTER
        );

        title.setFont(new Font("Arial", Font.BOLD, 25));
        title.setForeground(Color.WHITE);
        title.setOpaque(true);
        title.setBackground(new Color(40, 70, 120));
        title.setBorder(BorderFactory.createEmptyBorder(15, 10, 15, 10));

        add(title, BorderLayout.NORTH);

        // ---------- INPUT PANEL ----------
        JPanel inputPanel = new JPanel(new GridLayout(2, 4, 10, 10));
        inputPanel.setBorder(
            BorderFactory.createTitledBorder("Attendance Entry")
        );

        inputPanel.add(new JLabel("Student ID"));
        idField = new JTextField();
        inputPanel.add(idField);

        inputPanel.add(new JLabel("Student Name"));
        nameField = new JTextField();
        inputPanel.add(nameField);

        inputPanel.add(new JLabel("Date"));
        dateField = new JTextField("20-08-2026");
        inputPanel.add(dateField);

        inputPanel.add(new JLabel("Status"));
        statusBox = new JComboBox<String>(
            new String[]{"Present", "Absent"}
        );
        inputPanel.add(statusBox);

        // ---------- BUTTON PANEL ----------
        JPanel buttonPanel = new JPanel(
            new FlowLayout(FlowLayout.CENTER, 12, 8)
        );

        markBtn = new JButton("Mark Attendance");
        updateBtn = new JButton("Update");
        reportBtn = new JButton("View Report");
        clearBtn = new JButton("Clear");

        buttonPanel.add(markBtn);
        buttonPanel.add(updateBtn);
        buttonPanel.add(reportBtn);
        buttonPanel.add(clearBtn);

        // ---------- TOP PANEL ----------
        JPanel topPanel = new JPanel(new BorderLayout());
        topPanel.add(inputPanel, BorderLayout.CENTER);
        topPanel.add(buttonPanel, BorderLayout.SOUTH);

        add(topPanel, BorderLayout.CENTER);

        // ---------- TABLE ----------
        String[] columns = {
            "Student ID",
            "Student Name",
            "Date",
            "Status"
        };

        model = new DefaultTableModel(columns, 0) {

            // Prevent direct editing of table cells
            public boolean isCellEditable(int row, int column) {
                return false;
            }
        };
```

```
        table = new JTable(model);
        table.setRowHeight(28);
        table.getTableHeader().setFont(
            new Font("Arial", Font.BOLD, 14)
        );

        JScrollPane scrollPane = new JScrollPane(table);

        JPanel tablePanel = new JPanel(new BorderLayout());
        tablePanel.setBorder(
            BorderFactory.createTitledBorder("Attendance Records")
        );

        tablePanel.add(scrollPane, BorderLayout.CENTER);

        add(tablePanel, BorderLayout.SOUTH);

        // Make table larger
        tablePanel.setPreferredSize(new Dimension(900, 260));

        // ---------- EVENT HANDLING ----------
        markBtn.addActionListener(this);
        updateBtn.addActionListener(this);
        reportBtn.addActionListener(this);
        clearBtn.addActionListener(this);

        // Select table row when clicked
        table.addMouseListener(new MouseAdapter() {
            public void mouseClicked(MouseEvent e) {

                int row = table.getSelectedRow();

                if (row >= 0) {
                    idField.setText(
                        model.getValueAt(row, 0).toString()
                    );

                    nameField.setText(
                        model.getValueAt(row, 1).toString()
                    );

                    dateField.setText(
                        model.getValueAt(row, 2).toString()
                    );

                    statusBox.setSelectedItem(
                        model.getValueAt(row, 3).toString()
                    );
                }
            }
        });
    }

    // ---------- EVENT HANDLER ----------
    public void actionPerformed(ActionEvent e) {

        // Mark attendance
        if (e.getSource() == markBtn) {

            String id = idField.getText().trim();
            String name = nameField.getText().trim();
            String date = dateField.getText().trim();
            String status = statusBox.getSelectedItem().toString();

            // Validate empty fields
            if (id.equals("") || name.equals("") || date.equals("")) {
                JOptionPane.showMessageDialog(
                    this,
                    "Please enter all details!",
                    "Validation Error",
                    JOptionPane.ERROR_MESSAGE
                );
                return;
            }

            // Validate ID
            if (!id.matches("[0-9]+")) {
                JOptionPane.showMessageDialog(
                    this,
                    "Student ID must contain numbers only!",
                    "Invalid Data",
                    JOptionPane.ERROR_MESSAGE
                );
                return;
            }

            // Validate duplicate
            if (attendanceIds.contains(id + "_" + date)) {

                JOptionPane.showMessageDialog(
                    this,
                    "Attendance already marked for this student on this date!",
                    "Duplicate Entry",
                    JOptionPane.WARNING_MESSAGE
                );

                return;
            }

```

```
            // Add record
            model.addRow(
                new Object[]{id, name, date, status}
            );

            attendanceIds.add(id + "_" + date);

            JOptionPane.showMessageDialog(
                this,
                "Attendance marked successfully!"
            );

            clearFields();
        }

        // Update record
        else if (e.getSource() == updateBtn) {

            int row = table.getSelectedRow();

            if (row == -1) {
                JOptionPane.showMessageDialog(
                    this,
                    "Select a record to update!"
                );
                return;
            }

            String id = idField.getText().trim();
            String name = nameField.getText().trim();
            String date = dateField.getText().trim();
            String status = statusBox.getSelectedItem().toString();

            if (id.equals("") || name.equals("") || date.equals("")) {
                JOptionPane.showMessageDialog(
                    this,
                    "All fields are required!"
                );
                return;
            }

            model.setValueAt(id, row, 0);
            model.setValueAt(name, row, 1);
            model.setValueAt(date, row, 2);
            model.setValueAt(status, row, 3);

            JOptionPane.showMessageDialog(
                this,
                "Attendance record updated!"
            );

            clearFields();
        }

        // View report
        else if (e.getSource() == reportBtn) {

            int total = model.getRowCount();
            int present = 0;
            int absent = 0;

            for (int i = 0; i < total; i++) {

                String status =
                    model.getValueAt(i, 3).toString();

                if (status.equals("Present"))
                    present++;
                else
                    absent++;
            }

            JOptionPane.showMessageDialog(
                this,
                "ATTENDANCE REPORT\n\n" +
                "Total Records : " + total +
                "\nPresent       : " + present +
                "\nAbsent        : " + absent,
                "Attendance Report",
                JOptionPane.INFORMATION_MESSAGE
            );
        }

        // Clear fields
        else if (e.getSource() == clearBtn) {
            clearFields();
        }
    }

    // Clear input fields
    void clearFields() {
        idField.setText("");
        nameField.setText("");
        dateField.setText("20-08-2026");
        statusBox.setSelectedIndex(0);
        table.clearSelection();
    }
}
```

Output:


Applet Viewer: SmartAttendance.class
Applet
SMART ATTENDANCE SYSTEM
Attendance Entry
Student ID
124
Student Name
Gokul
Date
20-08-2026
Status
Present
Mark Attendance
Update
View Report
Clear
Attendance Records
Student ID
Student Name
Date
Status
123
Sathya
20-08-2026
Present
121
Krishna
20-08-2026
Absent
Applet started.


Applet Viewer: SmartAttendance.class
Applet
SMART ATTENDANCE SYSTEM
Attendance Entry
Student ID
Student Name
Date
20-08-2026
Present
Clear
Attendance Report
ATTENDANCE REPORT
Total Records : 3
Present : 2
Absent : 1
OK
Attendance Records
Student ID
Status
123
Sathya
Present
121
Krishna
20-08-2026
Absent
124
Gokul
20-08-2026
Present
Applet started.